

Ratio Problem Solving

Ratio mixed with fractions, percentages and changes: draw the parts, fill in what you know.



DR ESLAM AHMED
eliteigcse.com

THE TOPIC ON ONE MAP

Ratio Problem Solving

01 Ratio with fractions and percentages

$\frac{3}{10}$ into euros, rest 9 : 5 : 2

30% female

Joe gets 35%

02 Combining two or three ratios

almonds : cashews : peanuts

$e : g$ from $e : f, f : g$

$AB : BC : CD$

03 Before and after

each gives £10 to Pau

rent 2018 vs 2019

counters added

04 Ratio with algebra

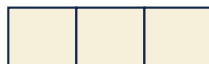
ages: 7 older, twice as old

$5a = 4c$

$(y + x) : (y - x) = k : 1$

BAR MODEL Draw the parts — every ratio problem becomes counting boxes:

Danil 3



Gabriel 5



Hadley 9



4 parts = 196 → 1 part = 49

01 Ratio with Fractions and Percentages

Change every fraction or percentage into parts of the same whole. What is left after a fraction is shared in a ratio — do the fraction first.

WHEN YOU SEE → YOU DO

“ $\frac{3}{10}$ into euros, the rest in 9 : 5 : 2”

→ rest = $\frac{7}{10}$, then split into 16 parts

“30% are female, 1295 are male”

→ male = 70%: 1% = 18.5

“Joe gets 35% in the ratio 7 : n”

→ 7 parts = 35%

“ $\frac{2}{3}$... $\frac{2}{5}$ of those ...”

→ multiply the fractions

WORKED MODEL • JANUARY 2021 P1H

20 160 rand: $\frac{3}{10}$ to euros, the rest in dollars : rupees : francs = 9 : 5 : 2.
How many more into dollars than francs?

1 rest = $\frac{7}{10} \times 20\,160 = 14\,112$

fraction first

2 $14\,112 \div 16 = 882$ per part

Zone A

ANSWER $(9 - 2) \times 882 = 6174$ rand

IDEA 01

Practise: Ratio with Fractions and Percentages

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Fraction first

CH 1 • TOPIC 13 3 marks

£600: $\frac{1}{4}$ is saved, the rest is shared by Ali and Ben in the ratio 2 : 7. How much does Ben get?

SELECTED PRACTICE

Q2 **PRACTICE** Practice • Medium

CH 1 • TOPIC 13 3 marks

- 2 30% of the people at a concert are female.
1295 of the people at the concert are male.

Work out the number of people at the concert who are female.

(3 marks)

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 13 **3 marks**

8 Anna and Lionel share \$675 in the ratio 4 : 5

Lionel gives $\frac{3}{5}$ of his share of the money to his mother.

How much money does Lionel give to his mother?

\$.....

(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 1 · TOPIC 13

4 marks

11 A bag contains some counters.

- There are 300 counters in the bag.
- There are only red, white and blue counters in the bag.
- The probability of picking a blue counter is $\frac{23}{50}$
- The ratio of red counters to white counters is 2 : 1.

Calculate the number of red counters in the bag.

(4 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 13 5 marks

10 The fares for a train journey are shown in the table below.

From London to Marseille	Standard fare	Premier fare
Adult	\$84	\$140
Child	\$60	\$96

For one journey from London to Marseille, the ratio

number of adults : number of children = 11 : 2.

There were 220 adults in total on this journey.

All of the children and 70% of the adults paid the standard fare.

The remaining adults paid the premier fare.

Calculate the total of the fares paid by the adults and the children.

\$

(5 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q1

CH 1 • TOPIC 13

3 marks

You must write down all the stages in your working.

- 1** Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6
The **total** amount of money that Asha and Julie receive is \$36

Work out the amount of money that Brendon receives.

\$.....

(Total for Question 1 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q1

CH 1 · TOPIC 13 4 marks

You must write down all the stages in your working.

- 1** Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

..... rand

(Total for Question 1 is 4 marks)

Q8**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q2

CH 1 · TOPIC 13 3 marks

- 2 Danil, Gabriel and Hadley share some money in the ratios 3:5:9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(Total for Question 2 is 3 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q6

CH 1 · TOPIC 13 4 marks

- 6 Eli and Peta share \$275 in the ratio 2:3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

02 Combining Two or Three Ratios

Make the shared item the same number in both ratios (use the LCM), then read any pair from the combined ratio.

WHEN YOU SEE → YOU DO

“ $e : f = 2 : 3$ and $f : g = 5 : 4$... find $e : g$ ”

→ $f \rightarrow 15: 10 : 15 : 12, e : g = 5 : 6$

“what fraction are sherberts?”

→ its part over the total parts

“there are less than 100 pens”

→ the total is a multiple of the sum

“ $AB : BD = 1 : 5, AC : CD = 7 : 11$ ”

→ same total length: 6 and 18 → use 18

WORKED MODEL • CASHEWS

1025 nuts. Almonds : cashews = 1 : 3, cashews : peanuts = 5 : 7. How many cashews?

1 cashews → 15: 5 : 15 : 21

combine

2 $1025 \div 41 = 25$

1 part

ANSWER $15 \times 25 = 375$ cashews

IDEA 02

Practise: Combining Two or Three Ratios

My questions first, then selected practice.

Q10 DR ESLAM Combine

CH 1 • TOPIC 13 3 marks

$x : y = 3 : 4$ and $y : z = 6 : 7$. Find $x : z$ in its simplest form.

SELECTED PRACTICE**Q11** PRACTICE Practice • Medium

CH 1 • TOPIC 13 2 marks

1 In a box of pens, there are

three times as many red pens as green pens

and two times as many green pens as blue pens.

For the pens in the box, write down

the ratio of the number of red pens to the number of green pens to the number of blue pens.

(2 marks)

Q12**PRACTICE**

Practice · Medium

CH 1 · TOPIC 13

4 marks

12 There is a total of 250 men, women and children on a train.

The ratio of men to women is 4 : 5.

The ratio of women to children is 10 : 7.

How many men are on the train?

(4 marks)**Q13****PRACTICE**

Practice · Hard

CH 1 · TOPIC 13

3 marks

4 There are only blue pens, green pens and red pens in a box.

The ratio of the number of blue pens to the number of green pens is 2 : 5

The ratio of the number of green pens to the number of red pens is 4 : 1

There are less than 100 pens in the box.

What is the greatest possible number of red pens in the box?

(3 marks)

Q14 **PRACTICE** Practice · Hard

CH 1 · TOPIC 13 3 marks

10 $e : f = 2 : 3$ and $f : g = 5 : 4$

Work out $e : g$

Give your answer in its simplest form.

(3 marks)

Q15 **PRACTICE** Practice · Very Hard

CH 1 · TOPIC 13 3 marks

4 The points A , B , C and D lie in order on a straight line.

$$AB : BD = 1 : 5$$

$$AC : CD = 7 : 11$$

Work out $AB : BC : CD$

(3 marks)

Q16

PRACTICE

Practice · Very Hard

CH 1 · TOPIC 136 marks

11 Some people who work for a company are in the following age groups.

Group A	Group B	Group C
Under 30 years	30 to 50 years	Over 50 years

The ratio of the number in group A to the number in group B is 7 : 10. The ratio of the number in group B to the number in group C is 4 : 3.

i) Find the ratio of the number in group A to the number in group C. Give your answer in its simplest form.

..... : [3]

ii) There are 45 people in group C.

Find the total number of people who work for the company.

[3]
(6 marks)

03 Before and After

Something changes — money given away, counters added. Work out the actual amounts before, apply the change, then write the new ratio. Look for what stays the same.

WHEN YOU SEE → YOU DO

“Sam and Tia each give £10 to Pau”

→ shares first, then +20, -10, -10

“Arjun paid the same rent both years”

→ make Arjun’s parts equal in both ratios

“she adds red counters, now 1 : 2”

→ yellow does not change

“the ratio is now 10 : 11 — more than 90?”

→ totals are multiples of 21

Amounts before

Apply the change

New ratio

Simplify

WORKED MODEL • JUNE 2025 P2H

Pau, Sam, Tia share £240 as 3 : 4 : 5. Sam and Tia each give £10 to Pau. New ratio?

1 $240 \div 12 = 20$: 60, 80, 100

amounts before

2 80, 70, 90

apply the change

ANSWER 8 : 7 : 9

IDEA 03

Practise: Before and After

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Before and after

CH 1 • TOPIC 13 3 marks

Red : blue counters = 2 : 5, 35 blue. 5 red are added. Find the new ratio of red : blue.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 13 4 marks

8 Arjun and Gretal each pay rent.

In 2018, the ratio of the amount each paid in rent was Arjun : Gretal = 5 : 7.

In 2019, the ratio of the amount each paid in rent was Arjun : Gretal = 9 : 13.

Arjun paid the same amount of rent in both 2018 and 2019.

Gretal paid \$290 more rent in 2019 than she did in 2018.

Work out the amount Arjun paid in rent in 2019.

\$

(4 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q6

CH 1 • TOPIC 13**4 marks**

- 6 Pau, Sam and Tia share £240 in the ratios 3:4:5

Sam and Tia each give £10 of their share to Pau.

Work out the ratios of the amounts of money that Pau, Sam and Tia now have.
Give your answer in its simplest form.

..... : :

(Total for Question 6 is 4 marks)

04 Ratio with Algebra

When the information is a sentence (“7 years older”, “twice as old”) or an equation ($5a = 4c$), write each quantity in terms of one letter, then form the ratio.

WHEN YOU SEE → YOU DO

“Kiaria is 7 years older than Jay; Martha is twice Kiaria”

→ $J, J + 7, 2(J + 7)$

“ b is two thirds of c ; $5a = 4c$ ”

→ choose c so all are whole: $c = 15$

“ $(y + x) : (y - x) = k : 1$ — show that ...”

→ $y + x = k(y - x)$, rearrange

WORKED MODEL • AGES

Kiaria is 7 years older than Jay, Martha is twice as old as Kiaria, sum 77. Find Jay : Kiaria : Martha.

1 $J + (J + 7) + 2(J + 7) = 77 \Rightarrow 4J = 56$

one letter

2 $14 : 21 : 42$

ages

ANSWER $2 : 3 : 6$

IDEA 04

Practise: Ratio with Algebra

My questions first, then selected practice.

Q20 **DR ESLAM** One letter

CH 1 • TOPIC 13 3 marks

Amy is 4 years older than Bea. Cara is 3 times Bea's age. Total 49. Find Amy : Bea : Cara.

SELECTED PRACTICE**Q21** **PRACTICE** Practice • Hard

CH 1 • TOPIC 13 4 marks

- 3 Kiaria is 7 years older than Jay.
Martha is twice as old as Kiaria.
The sum of their three ages is 77

Find the ratio of Jay's age to Kiaria's age to Martha's age.

(4 marks)

Q22**PRACTICE**

Practice · Very Hard

CH 1 · TOPIC 13 3 marks

6 b is two thirds of c .

$$5a = 4c$$

Work out the ratio $a : b : c$

Give your answer in its simplest form where a , b and c are integers.

(3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Fractions / % | Everything as parts of one whole; fraction first. |
| 02 | Combine | Shared item → LCM; read any pair. |
| 03 | Before / after | Real amounts, change them, new ratio — find what stays fixed. |
| 04 | Algebra | One letter for everything, then the ratio. |

BOTTOM LINE Draw the parts → value of 1 part → amounts → then the change or the fraction.

Answers

Ratio Problem Solving • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RATIO WITH FRACTIONS AND PERCENTAGES

Q1	Dr Eslam • Fraction first	rest £450: $450 \div 9 \times 7 = \text{£}350$
Q2	Practice • Medium	555
Q3	Practice • Medium	\$225
Q4	Practice • Hard	108
Q5	Practice • Very Hard	\$24 576
Q6	Edexcel 4MA1 • January 2020 • Paper 1HR • Q1	\$13.50
Q7	Edexcel 4MA1 • January 2021 • Paper 1H • Q1	6174 rand
Q8	Edexcel 4MA1 • January 2022 • Paper 1H • Q2	147 euros
Q9	Edexcel 4MA1 • November 2025 • Paper 1H • Q6	$30 + 52.80 = \$82.80$

IDEA 02 • COMBINING TWO OR THREE RATIOS

Q10	Dr Eslam • Combine	$9 : 12 : 14$; $x : z = 9 : 14$
Q11	Practice • Medium	$6 : 2 : 1$
Q12	Practice • Medium	80
Q13	Practice • Hard	15
Q14	Practice • Hard	$5 : 6$
Q15	Practice • Very Hard	$3 : 4 : 11$
Q16	Practice • Very Hard	(i) $14 : 15$ (ii) 147

IDEA 03 • BEFORE AND AFTER

Q17	Dr Eslam • Before and after	$14 \text{ red} \rightarrow 19$; $19 : 35$
Q18	Practice • Very Hard	\$6525
Q19	Edexcel 4MA1 • June 2025 • Paper 2H • Q6	$8 : 7 : 9$

IDEA 04 • RATIO WITH ALGEBRA

Q20	Dr Eslam • One letter	$5B + 4 = 49$, $B = 9$; $13 : 9 : 27$
Q21	Practice • Hard	$2 : 3 : 6$
Q22	Practice • Very Hard	$12 : 10 : 15$